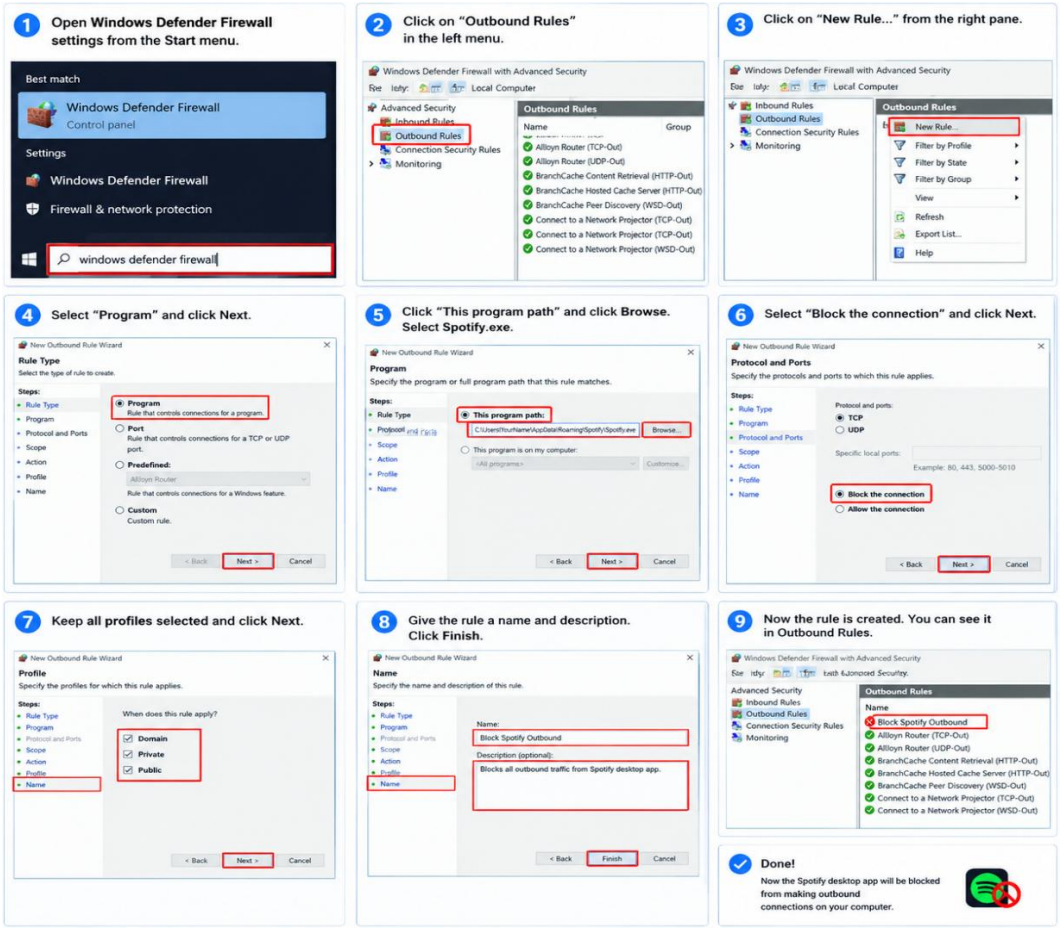
Day - 4
Firewall Basics

Q1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Answer:

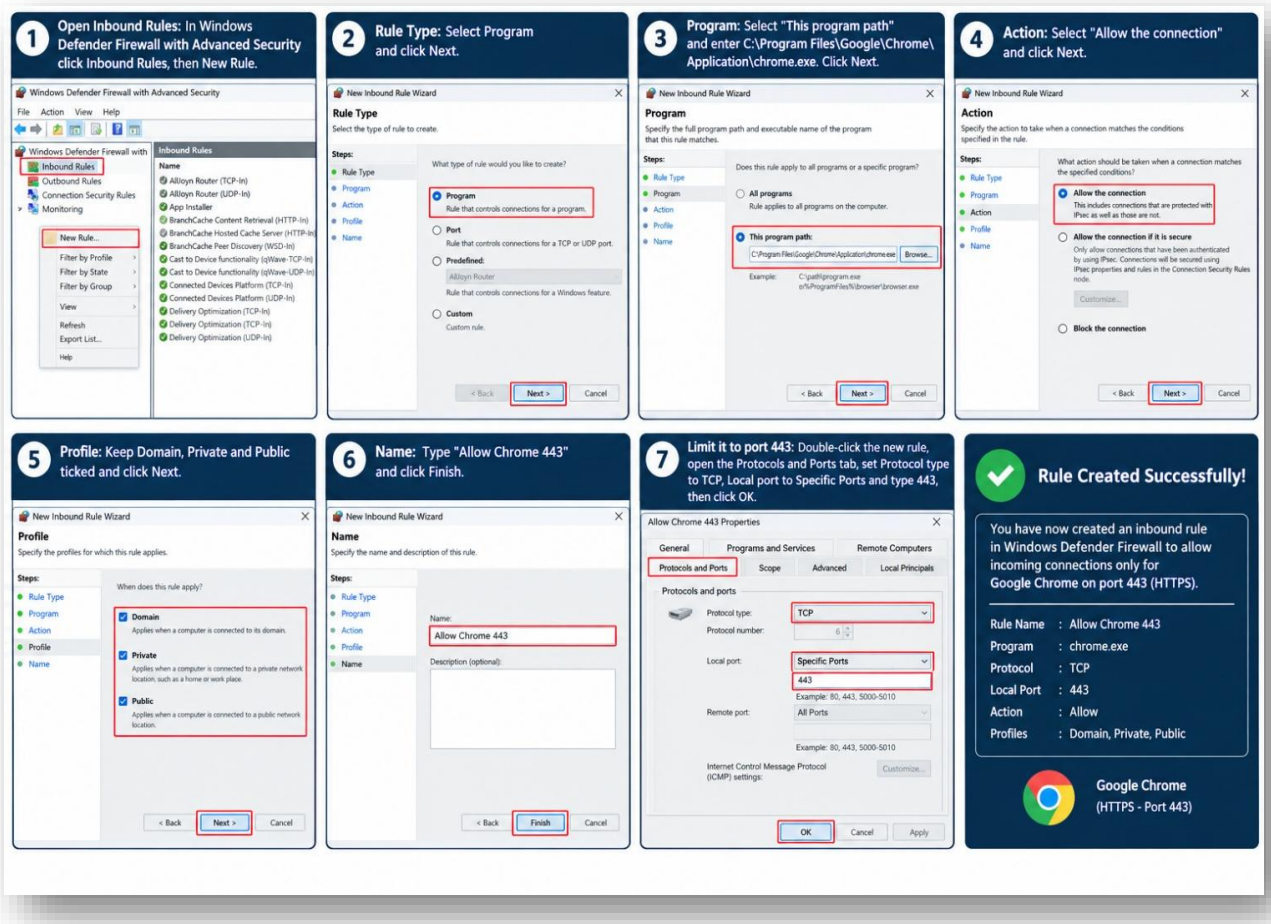
1. **Open the firewall:** Click Start, type "Windows Defender Firewall with Advanced Security" and open it.
2. **Choose Outbound Rules:** In the left pane click Outbound Rules, then in the right pane click New Rule.
3. **Rule Type:** Select Program and click Next.
4. **Program:** Select "This program path" and enter the Spotify path, for example C:\Users\<username>\AppData\Roaming\Spotify\Spotify.exe (use Browse to find it). Click Next.
5. **Action:** Select "Block the connection" and click Next.
6. **Profile:** Keep Domain, Private and Public ticked and click Next.
7. **Name:** Type "Block Spotify" and click Finish.
8. **Test:** Open Spotify. It cannot connect to the Internet any more (songs do not load). To allow it again, delete or disable the rule.



Q2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Answer:

9. **Open Inbound Rules:** In Windows Defender Firewall with Advanced Security click Inbound Rules, then New Rule.
10. **Rule Type:** Select Program and click Next.
11. **Program:** Select "This program path" and enter C:\Program Files\Google\Chrome\Application\chrome.exe. Click Next.
12. **Action:** Select "Allow the connection" and click Next.
13. **Profile:** Keep Domain, Private and Public ticked and click Next.
14. **Name:** Type "Allow Chrome 443" and click Finish.
15. **Limit it to port 443:** Double-click the new rule, open the Protocols and Ports tab, set Protocol type to TCP, Local port to Specific Ports and type 443, then click OK.



Q3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Answer:

Traffic can come from the IP address (inbound) and go to it (outbound), so I create one rule for each direction. The steps for the inbound rule are:

1. **Open the firewall:** Open Windows Defender Firewall with Advanced Security.
2. **New rule:** Click Inbound Rules, then New Rule.
3. **Rule Type:** Select Custom and click Next.
4. **Program:** Select "All programs" and click Next.
5. **Protocol and Ports:** Keep Protocol type as Any and click Next.
6. **Scope:** Under "Which remote IP addresses does this rule apply to?" select "These IP addresses", click Add, type 123.45.67.89 and click OK, then Next. (This is the Scope setting from the hint.)
7. **Action:** Select "Block the connection" and click Next.
8. **Profile:** Keep Domain, Private and Public ticked and click Next.
9. **Name:** Type "Block IP 123.45.67.89 In" and click Finish.
10. **Outbound rule:** Repeat the same steps under Outbound Rules and name it "Block IP 123.45.67.89 Out".

Q4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Answer:

Point	Hardware firewall	Software firewall
Where it works	A separate device placed between the Internet and the network. It protects every computer behind it.	A program installed on one computer. It protects only that computer.
Performance	Has its own processor and memory, so it handles heavy traffic from many users without slowing computers.	Uses the computer's own CPU and RAM, so it can slightly slow the computer.
Cost and management	Costly, and needs trained staff, but is managed centrally for the whole network.	Cheap or free (Windows Defender comes with Windows), easy to set up, but must be set on each computer separately.
Use case	The gateway of a large office or college, protecting hundreds of computers.	A personal laptop at home or on public WiFi, to control which apps can connect.